

# Physics Questions - motion in 1 d include graphs also

Subject: Physics | Chapter: motion in 1 d include graphs also | Difficulty: medium

**Question 1:** A car accelerates uniformly from rest to 20 m/s in 10 seconds. What is its acceleration?

Options:

- A) 1 m/s<sup>2</sup>
- B) 2 m/s<sup>2</sup>
- C) 5 m/s<sup>2</sup>
- D) 10 m/s<sup>2</sup>

Difficulty: 2/5

**Question 2:** A ball is thrown vertically upwards with an initial velocity of 20 m/s. Ignoring air resistance, what is its velocity after 2 seconds? ( $g = 10 \text{ m/s}^2$ )

Options:

- A) 0 m/s
- B) 10 m/s
- C) 20 m/s
- D) 30 m/s

Difficulty: 3/5

**Question 3:** The graph shows the velocity-time graph of an object. What is the displacement of the object between  $t = 2\text{s}$  and  $t = 6\text{s}$ ? (Assume a linear relationship between points)

Options:

- A) 10m
- B) 20m
- C) 30m
- D) 40m

Difficulty: 3/5

**Question 4:** A body is moving with a constant velocity. Which of the following statements is true?

Options:

- A) Its acceleration is zero.
- B) Its displacement is zero.
- C) Its speed is changing.
- D) Its momentum is changing.

*Difficulty: 2/5*

**Question 5:** A stone is dropped from a height. Ignoring air resistance, which of the following is constant?

Options:

- A) Its velocity
- B) Its acceleration
- C) Its momentum
- D) Its kinetic energy

*Difficulty: 2/5*

**Question 6:** The area under a velocity-time graph represents:

Options:

- A) Velocity
- B) Acceleration
- C) Displacement
- D) Speed

*Difficulty: 2/5*

**Question 7:** A train travels 120 km in 2 hours and then 180 km in 3 hours. What is its average speed for the entire journey?

Options:

- A) 40 km/h
- B) 50 km/h
- C) 60 km/h
- D) 70 km/h

*Difficulty: 2/5*

**Question 8:** A freely falling object experiences:

Options:

- A) constant velocity
- B) increasing acceleration
- C) constant acceleration
- D) decreasing acceleration

*Difficulty: 2/5*

**Question 9:** Which equation of motion is used to find the final velocity when initial velocity, acceleration and displacement are given?

Options:

- A)  $v = u + at$
- B)  $s = ut + (1/2)at^2$
- C)  $v^2 = u^2 + 2as$
- D)  $s = (u+v)t/2$

*Difficulty: 3/5*

**Question 10:** A ball is thrown horizontally from a cliff. Ignoring air resistance, which statement is true about its vertical motion?

Options:

- A) It has constant velocity.
- B) It has zero acceleration.
- C) It has constant acceleration downwards.
- D) It has constant acceleration upwards.

*Difficulty: 3/5*

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